Johannes Hollenbach

johanneshollenbach.com | +49 201 8149 261 | johannes.hollenbach@rwi-essen.de

Research Experience

Research Associate, Chair of Empirical Economics, Paderborn University 2024 – present

Research Associate, RWI – Leibniz Institute for Economic Research 2021 – present

Research Fields

Primary: Health Economics
Secondary: Labor and Education Economics

Education

Ph.D. in Economics (Dr. rer. pol.), Paderborn University 2022 – present
Dissertation: How Endowments and Policies Shape Major Life Decisions and Healthy Aging
Advisor: Hendrik Schmitz

Visiting Scholar, University of Wisconsin–Madison Fall 2024
La Follette School of Public Affairs, *Host*: Jason Fletcher

Visiting Scholar, Vienna University of Economics and Business Fall 2021
Institute for Social Policy, *Host*: Sophie Guthmuller

M.Sc. in Economics, University of Bayreuth 2021

B.A. in International Economics and Development, University of Bayreuth 2018

Job Market Paper

Genetic Endowments and the Long-Run Health Returns to Compulsory Schooling

Genetic differences tied to educational attainment predict later-life success and wellbeing. I study whether they also moderate the causal effect of education on physical health in old age. Using data on individuals aged 65 to 85 from the English Longitudinal Study of Ageing, I exploit the 1947 raising of the school leaving age in the UK from 14 to 15 in a fuzzy regression discontinuity setting to estimate the causal effect of schooling on frailty, the share of age-related health deficits. I estimate interactions with an educational attainment polygenic index (EA PGI), a DNA-based measure of genetic propensity for education. Health returns to schooling are concentrated at the top of the EA PGI distribution. In the highest quintile, additional education reduces frailty by 6 percentage points, compared with near-zero effects at the bottom and possibly small benefits in the middle. This suggests a complementarity between genetic endowments and education. These health improvements originate at the upper part of the frailty distribution. I further show that the EA PGI also affects educational responses to the schooling reform. Using marginal treatment effects, I find that the main results are not driven by these differences in compliance behavior or unobserved characteristics of compliers.

Publications

Gene-Environment Interactions with Essential Heterogeneity (with H. Schmitz, M. Westphal). ***The Economic Journal***, forthcoming.

Evolution of Digital Maturity of German Hospitals: A Longitudinal Analysis from 2021 to 2024 (with M. Haring, F. Hasse, A. Haering, K. Materne, S. Rohwer, F. Fritzsche, S. Thun, V. E. Amelung, J. Peukert, A. Geissler). ***Health Policy and Technology***, forthcoming.

Life-Cycle Health Effects of Compulsory Schooling (with H. Schmitz, B. B. Tawiah). ***The European Journal of Health Economics***, 27(6), 2026.

The Impact of Retirement on Loneliness in Europe (with S. Guthmuller, D. Heger, A. Werbeck). ***Scientific Reports***, 14(1), 2024.

A Nationwide Digital Maturity Assessment of Hospitals: Results from the German DigitalRadar (with A. Geissler, M. Haring, V. E. Amelung, S. Thun, A. Haering). ***Health Policy and Technology***, 13(4), 2024.

The Association of Hospital Profitability and Digital Maturity: An Explorative Study Using Data from the German DigitalRadar Project (with J. Vogel, A. Haering, B. Augurzky, A. Geissler). ***Health Policy***, 142, 2024.

Awards & Grants

Research Exchange Scholarship, RWI – Leibniz Institute for Economic Research 2024

Travel Grant, German Academic Exchange Service (DAAD) 2024

Invited Talks

Health Insurance Committee Meeting, German Health Economics Association 2026

Applied Micro Workshop, University of Wisconsin–Milwaukee 2024

Initiative in Social Genomics, University of Wisconsin–Madison (online) 2024

Conferences & Workshops

2026 German Health Economics Association Annual Meeting (Wuppertal), European Social Science Genetics Network Conference (Bologna), Leibniz Resilient Ageing Workshop (Halle)

2025 RWI Brown Bag (Essen), German Health Economics Association Annual Meeting (Paderborn), University of Duisburg-Essen Micro Workshop (Kamen)

2024 ASSA Annual Meeting (San Antonio), Initiative in Social Genomics (UW–Madison), European Health Economics Association PhD & Supervisor Conference (Lucerne), CINCH-dggö Academy in Health Economics (Essen), German Health Economics Association Annual Meeting (Halle), Center for Economic and Statistical Analysis Young Researcher Workshop (Hagen)

2023 European Health Economics Association PhD & Supervisor Conference (Bologna), CINCH-dggö Academy in Health Economics (Essen)

2022 European Health Economics Association PhD & Supervisor Conference (Galway), European Health Economics Association Conference (Oslo), Economics of Mental Health Workshop (Essen), German Health Economics Association Annual Meeting (Hamburg), RWI Brown Bag (Essen)

Refereeing

Health Economics, *Health Policy and Technology*, *The B.E. Journal of Economic Analysis & Policy*, *The European Journal of Health Economics*

Outreach & Service

Transfer Manager, Leibniz ScienceCampus "Skills over the Lifecycle" 2026 – present

PhD Representative (elected), RWI – Leibniz Institute for Economic Research 2025 – present

Scientific Partner, Young Economic Solutions 2022, 2023

Teaching Experience

M.Sc. Thesis Supervision, Paderborn University 2024 – present

Teaching Assistant, Paderborn University
- Statistics for Business Informatics (undergraduate) 2025
- Microeconometrics (graduate) 2024, 2025
- Empirical Economics (undergraduate) 2024

Teaching Assistant, University of Duisburg-Essen
- Healthcare Systems (graduate) 2024

Further Research Experience

Student Assistant, Chair of Development Economics, University of Bayreuth 2018 – 2021

Student Assistant, Chair of Institutional Economics, University of Bayreuth 2018 – 2021

Affiliations & Memberships

American Economic Association (AEA), German Health Economics Association (dggö)

References

Hendrik Schmitz
Paderborn University
hendrik.schmitz@uni-paderborn.de

Matthias Westphal
FernUniversität in Hagen
matthias.westphal@fernuni-hagen.de

Ansgar Wübker
Hochschule Harz
awuebker@hs-harz.de

Last updated: November 2026